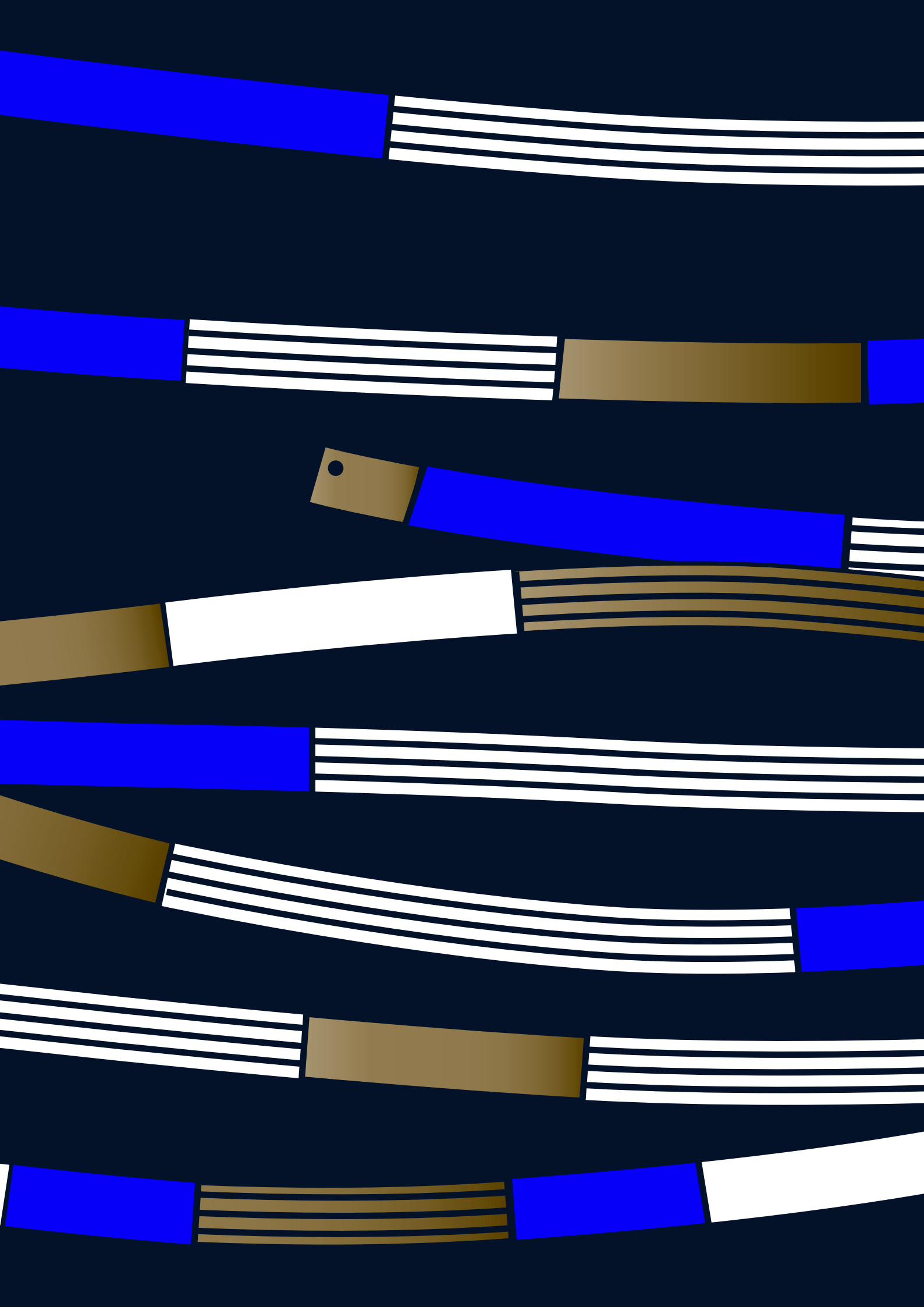
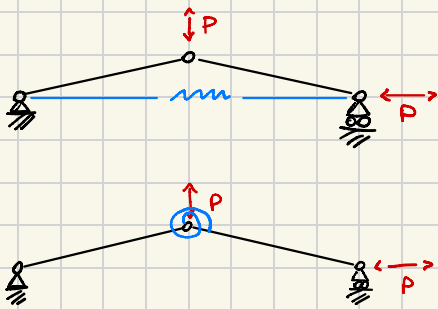




Algorithm:

- 1) Write: $V(\theta) = V_{el}(\theta) - W_{ext}(\theta)$
 - 2) Find 1st and 2nd derivative
 - 3) Try to draw 1st derivative and study the number of solution ($F/P \neq 0$ and $F/P = 0$)
 - 4) Do the Taylor expansion (Approximate P/F respect to the angle)
 - 5) Study also the sign of $\frac{d^2V(\theta)}{d\theta^2} \rightarrow \begin{cases} > , \text{ stable} \\ = 0 , \theta_{cr} \sim P/F_{limit} \\ < , \text{ unstable} \end{cases}$
-

Type & structure:



Pitchwork bifurcation

PERCORSO DIRAMATO STABILE (same as professor)

Si consideri una trave rigida con carico di compressione P a una estremità e cerniera elastica all'estremità opposta, Fig. 23.4a.

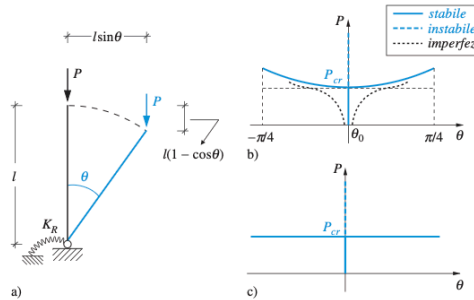
La configurazione del sistema è completamente individuata dall'unico parametro θ (coordinata lagrangiana) e possiede dunque un unico grado di libertà; l'asta è soggetta alla forza P e al momento di richiamo della molla $M_R = -K_R\theta$. La posizione iniziale con l'asta verticale è di equilibrio e, a norma del criterio di Eulero, il valore critico (P_{cr}) per il carico P si ottiene quando sono possibili posizioni di equilibrio distinte da questa. Si scriva allora l'equazione di equilibrio dei momenti nella configurazione deformata in spostamenti finiti. Scelto come polo la cerniera, si ha:

$$Pl\sin\theta - K_R\theta = 0 \quad \Rightarrow \quad Pl\sin\theta = K_R\theta \quad (23.1)$$

Dalla precedente relazione si vede che le posizioni di equilibrio per il sistema sono:

- 1) $\theta = 0$, P qualsiasi (Percorso fondamentale);
- 2) $\theta \neq 0$, $P = \frac{K_R}{l} \frac{\theta}{\sin\theta}$ (Percorso diramato). (23.2)

Figura 23.4
Asta rigida compressa con cerniera elastica:
a) configurazione iniziale e variata;
b) diagramma di biforcazione in regime di spostamenti finiti;
c) diagramma di biforcazione in regime di piccoli spostamenti



Queste soluzioni, riportate nel piano $\theta - P$, formano un diagramma detto di *biforcazione*, Fig. 23.4b: poiché per $P < \frac{K_R}{l}$ si ha una sola soluzione (stabile) e per $P > \frac{K_R}{l}$ se ne hanno tre (due stabili e una instabile), ne consegue che, in accordo con il criterio di Eulero, il carico critico è $P_{cr} = \frac{K_R}{l}$. Il punto del diagramma corrispondente al carico critico in cui si ha perdita

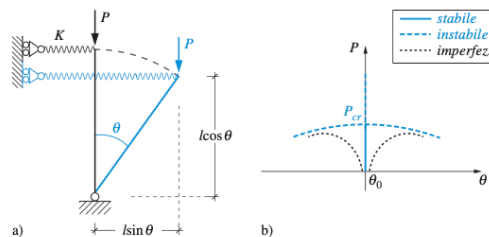
dell'unicità della soluzione viene detto *punto di biforcazione*. Utilizzando criteri energetici è possibile valutare la stabilità dei percorsi di equilibrio come riportato nella stessa Figura 23.4b: si osservi che il percorso diramato è costituito da posizioni di equilibrio stabile e che, sul percorso fondamentale, l'equilibrio è stabile per $P < P_{cr}$ e instabile per $P > P_{cr}$.

Un'analisi semplificata si ottiene facendo l'ipotesi di piccoli spostamenti: $\theta \rightarrow 0$, $\frac{\sin\theta}{\theta} \rightarrow 1$. Il diagramma corrispondente è riportato Fig. 23.4c: come si vede il valore del carico critico è individuato correttamente ma si perdono buona parte delle informazioni sul comportamento post-critico.

PERCORSO DIRAMATO INSTABILE

Si consideri un'asta rigida incernierata con carico di compressione P e molla trasversale all'estremo libero, Fig. 23.5a.

Come nel caso precedente, la configurazione del sistema è completamente individuata dall'unico parametro θ . L'asta è soggetta al carico P e alla forza di richiamo della molla $F_{el} = -K\Delta x = -Kl\sin\theta$. Scelto come polo la cerniera, l'equazione di equilibrio dei momenti nella configurazione deformata in spostamenti finiti si scrive:



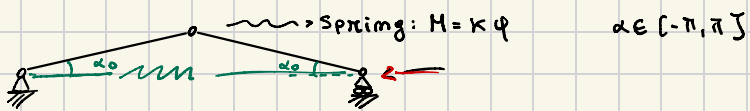
$$Pl\sin\theta - Kl\sin\theta \cdot \cos\theta = 0 \quad (23.3)$$

Dalla precedente relazione si deduce che sono posizioni di equilibrio per il sistema le seguenti:

- 1) $\theta = 0$, P qualsiasi (Percorso fondamentale);
- 2) $\theta \neq 0$, $P = Kl \cos\theta$ (Percorso diramato). (23.4)

La stabilità dei vari percorsi di equilibrio, valutata con criteri energetici, è riportata in Fig. 23.5b. Ragionando come prima, si deduce che il carico critico vale in questo caso: $P_{cr} = Kl$. Rispetto al precedente esempio, si nota che il percorso diramato è instabile e decrescente: ciò ha una profonda influenza sulla risposta del sistema in presenza di imperfezioni come mostra il prossimo paragrafo.

Figura 23.5
Asta rigida compressa vincolata elasticamente:
a) configurazione iniziale e variata;
b) diagramma di biforcazione in regime di spostamenti finiti



$$V(\alpha) = \frac{1}{2}k \left((\pi - 2\alpha) - (\pi - 2\alpha_0) \right)^2 - \left(Pl(2\cos\alpha - 2\cos\alpha_0) \right)$$

$$= \frac{1}{2}k \cdot 4(\alpha_0 - \alpha)^2 - 2Pl(\cos\alpha - \cos\alpha_0)$$

$$= 2k(\alpha_0 - \alpha)^2 - 2Pl(\cos\alpha - \cos\alpha_0) \quad \text{or}$$

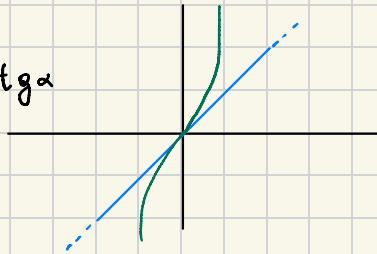
$$\frac{dV}{d\alpha} = 4k(\alpha - \alpha_0) + 2Pl\sin\alpha$$

$$P(\alpha) = 2 \frac{k}{l} \left(\frac{\alpha_0 - \alpha}{\sin\alpha} \right) \rightsquigarrow \begin{cases} \alpha_0 = 0, P_{cr} = -2 \frac{k}{l} (?) \\ P = 0, \alpha = \alpha_0 \end{cases}$$

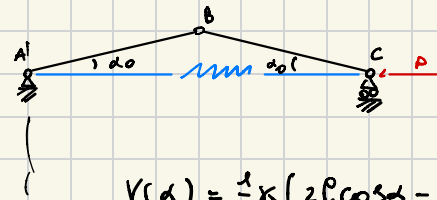
(idk Taylor)

$$\frac{dP}{d\alpha} = 2 \frac{k}{l} \left(\frac{-\sin\alpha + \alpha \cos\alpha}{\sin^2\alpha} \right) \overset{\text{for stability}}{\uparrow} < 0 \rightsquigarrow \text{just the num. } -\sin\alpha + \alpha \cos\alpha > 0 \iff \alpha < \text{tga}$$

$$\begin{cases} \alpha > 0 \text{ (stable)} \\ \alpha < 0 \text{ (unstable)} \end{cases}$$



$$P=0 \quad \alpha = \pm \alpha_0, \alpha = 0$$



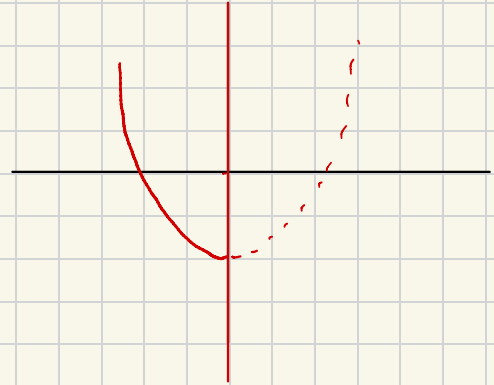
$$P = 2k\ell(\cos\alpha - \cos\alpha_0)$$

$$= 2k\ell \left(+\frac{\alpha^2}{2} - \frac{\alpha_0^2}{2} \right) \quad k\ell(\alpha^2 - \alpha_0^2)$$

$$V(\alpha) = \frac{1}{2}k(2\ell\cos\alpha - 2\ell\cos\alpha_0)^2 + P\ell(2\cos\alpha - 2\cos\alpha_0)$$

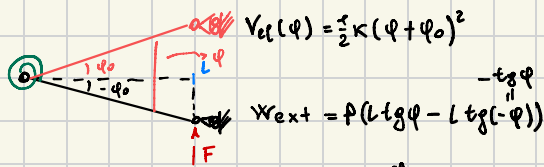
$$\frac{dV}{d\alpha} = 4k\ell^2(\cos\alpha - \cos\alpha_0)(-\sin\alpha) - \cancel{2P\ell\sin\alpha}$$

$$(4k\ell^2(\cos\alpha - \cos\alpha_0) + 2P\ell)$$



$$(-10 + \alpha_0^2)\alpha^2 = -8$$

$$\alpha^2 > -\frac{8}{\alpha^2 - 10}$$



$$V_{el}(\varphi) = \frac{1}{2} \kappa (\varphi + \varphi_0)^2$$

$$W_{ext} = P(l \lg \varphi - l \lg(-\varphi))$$

$$V(\alpha) = \frac{1}{2} \kappa (\varphi + \varphi_0)^2 - PL(\tilde{\lg} \varphi + l \lg \varphi_0)$$

$$\frac{dV}{d\varphi} = \kappa(\varphi + \varphi_0) - PL \sim \varphi = \frac{\kappa}{L}(\varphi + \varphi_0) \quad \begin{matrix} P=0 \\ P \neq 0 \end{matrix}$$

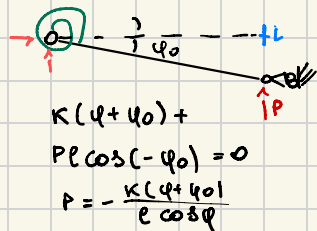
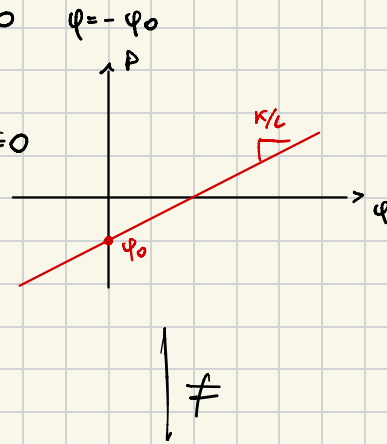
$$\frac{d^2V}{d\varphi^2} = \kappa \sim \kappa > 0 \quad \forall \varphi \text{ (always stable (?))}$$

(No Taylor)

$$V(\varphi) = \frac{1}{2} \kappa (\varphi + \varphi_0)^2 - PL \lg \varphi$$

$$\begin{aligned} \frac{dV}{d\varphi} &= \kappa(\varphi + \varphi_0) - PL \cdot \frac{1}{\cos^2 \varphi} \\ &= \kappa(\varphi + \varphi_0) - PL \left(1 + \frac{\varphi^2}{2}\right)^2 \end{aligned}$$

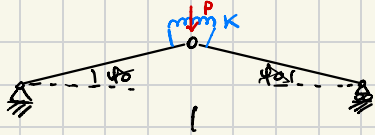
$$P = \frac{\kappa}{L} \cos^2 \varphi (\varphi + \varphi_0) \approx P = \frac{\kappa}{L} \left(1 - \frac{\varphi^2}{2}\right)^2 (\varphi + \varphi_0)$$



$$\kappa(\varphi + \varphi_0) +$$

$$PL \cos(-\varphi_0) = 0$$

$$P = -\frac{\kappa(\varphi + \varphi_0)}{L \cos \varphi}$$



$$\cos \varphi = -\frac{1}{2} \frac{k}{P} (\varphi - \varphi_0)$$

$$V(\varphi) = \frac{1}{2} k (\frac{\pi}{2} - \varphi) - (\frac{\pi}{2} - \varphi_0))^2 + P l (\sin \varphi - \sin \varphi_0)$$

$$P=0 \quad \varphi_0 = \varphi$$

$$\frac{dV}{d\varphi} = k(\varphi - \varphi_0) + P l \cos \varphi \Leftrightarrow P = \frac{k}{l} \frac{\varphi_0 - \varphi}{\cos \varphi}$$

$$P = \frac{k}{l} \frac{\varphi_0}{\cos \varphi_0}$$

$$\frac{d^2 V}{d\varphi^2} = k - P l \sin \varphi$$

$$= k - \frac{k}{l} (\varphi_0 - \varphi) \left(1 + \frac{\varphi^2}{2}\right)$$

$$= \frac{k}{l} \left(-1 - \frac{\varphi^2}{2} + \varphi \varphi_0 - \varphi^2\right)$$

$$1 - (\varphi_0 - \varphi) \tan \varphi > 0$$

$$\tan \varphi < \frac{1}{\varphi_0 - \varphi}$$

$$k - P l \sin \varphi$$

$$k - P l \varphi > 0$$

$$\varphi_0 \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\varphi < 0$$

$$-2 - 3\varphi^2 + 2\varphi\varphi_0 > 0$$

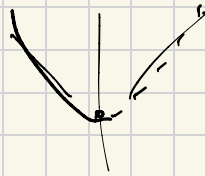
$$\varphi_0^2 > 2$$

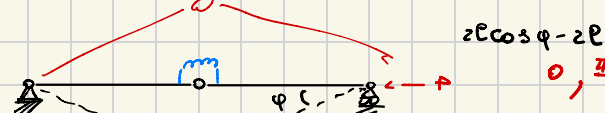
$$3\varphi^2 - 2\varphi\varphi_0 + 2 < 0$$

$$\varphi_0$$

$$4\varphi_0^2 - 8$$

$$\frac{2\varphi_0 \pm 2\sqrt{4\varphi_0^2 - 8}}{6}$$





$$V(x) = \frac{1}{2} k (z\varphi)^2 + \underbrace{PE(z \cos \varphi - z)}_{0, \pi} \quad \varphi \rightarrow 0$$

$$V'(x) = k\varphi - zPE \sin \varphi$$

$$\rightarrow P(x) = \frac{z k \varphi}{l \sin \varphi} = z \frac{k}{l} \cdot \frac{\varphi}{\sin \varphi} \sim z \frac{k}{l} \varphi$$

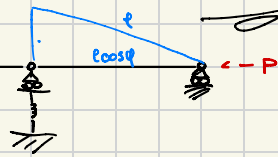
$$V''(x) = k - zPE \cos \varphi \quad \varphi < \frac{\pi}{2} \quad (?)$$

$$k - zPE \left(1 + \frac{\alpha^2}{2}\right)$$

$$\frac{z}{l} k - z k \varphi \cot \varphi > 0 \quad \varphi \cot \varphi < z$$

$$\frac{\varphi}{\varphi - \frac{\varphi^3}{6}} \quad 1 - \frac{\varphi^2}{6}$$

$$\cot \varphi < \frac{z}{\varphi} \quad \text{Semper } k \neq P(\varphi)$$



$$W_{exp} = -zPE(\cos \varphi - 1)$$

$$V(\varphi) = \frac{1}{2} k (l \sin \varphi)^2 + PE(z \cos \varphi - z)$$

$$\frac{dV}{d\varphi} = k l^2 \sin \varphi \cos \varphi - zPE \sin \varphi \quad \varphi = 0 \text{ equilibrium}$$

$$\sin \varphi (zPE - k l^2 \cos \varphi)$$

$$P = \frac{k}{2} l \cos \varphi$$

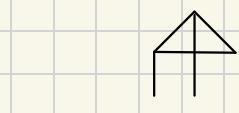
$$\frac{dV}{d\varphi} = \frac{1}{2} k l^2 \sin 2\varphi - zPE \sin \varphi$$

$$\frac{d^2V}{d\varphi^2} = k l^2 \cos 2\varphi - zPE \cos \varphi$$

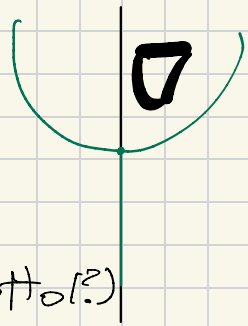
$$k l^2 \cos 2\varphi - k l^2 \cos 2\varphi$$

$$k l^2 (\cos 2\varphi - \sin 2\varphi) - k l^2 \sin 2\varphi < 0$$

Never

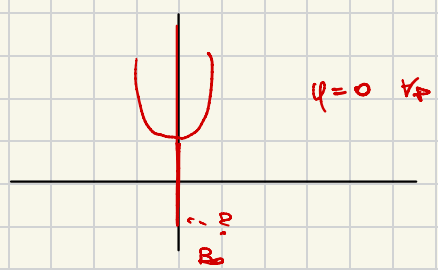


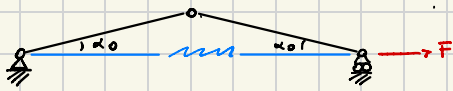
$$\frac{1}{\sin \varphi} \sim \frac{1}{\sin^3 \varphi}$$



$$F = z \frac{k}{l} \frac{\varphi}{\sin \varphi} \quad \varphi \rightarrow 0 \quad 1 \quad \varphi \rightarrow \pi/2$$

$$\varphi \rightarrow \frac{\pi}{2} \quad \varphi \rightarrow -\frac{\pi}{2}$$





$$V_{el}(\alpha) = \frac{1}{2} k (2l \cos \alpha - 2l \cos \alpha_0)^2$$

$$W_{ext} = P (2l \cos \alpha - 2l \cos \alpha_0)$$

$$V(\alpha) = \frac{1}{2} k (2l \cos \alpha - 2l \cos \alpha_0)^2 - 2Pl (\cos \alpha - \cos \alpha_0)$$

$$\frac{dV}{d\alpha} = 4k l^2 (\cos \alpha - \cos \alpha_0) (-\sin \alpha) + 2Pl \sin \alpha$$

$$= (\sin \alpha) (Pl - 2k l^2 (\cos \alpha - \cos \alpha_0))$$

$$P = 2k l^2 (\cos \alpha - \cos \alpha_0) \approx k l^2 (\alpha_0^2 - \alpha^2)$$

$$\frac{dV}{d\alpha} = k l^2 (-2 \sin 2\alpha + 2 \sin 2\alpha \cos \alpha_0) + Pl \sin \alpha$$

$$\frac{d^2V}{d\alpha^2} = k l^2 (-4 \cos 2\alpha + 2 \cos 2\alpha \cos \alpha_0) + Pl \cos \alpha$$

$$Pl - 4k l^2$$

$$2k l^2 \cos^2 \alpha - 2k l^2 \cos 2\alpha \cos \alpha_0$$

$$-4k l^2 \cos^2 \alpha + 2k l^2 \cos^2 \alpha$$

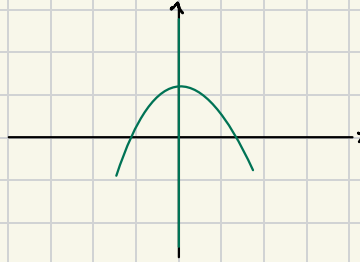
$$-4k l^2 \cos^2 \alpha - 4k l^2 \sin^2 \alpha + 2k l^2 \cos^2 \alpha$$

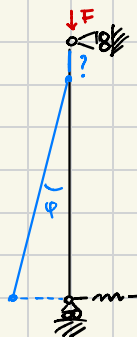
$$+ 2k l^2 \cos^2 \alpha + 2k l^2 \sin^2 \alpha < 0$$

$$1 + 2 \tan^2 \varphi < 0 \quad \text{stable} \quad \forall \varphi$$

$$\tan^2 \varphi < -\frac{1}{2}$$

$\alpha = 0 \quad \forall P$ equilibrium
 $P = 0 \quad \alpha = \pm \alpha_0$





$$\frac{1}{2} k (l \sin \varphi)^2 + P(l \cos \varphi - l)$$

$$\frac{dV}{d\varphi} = k l^2 \cos \varphi \sin \varphi - P l \sin \varphi$$

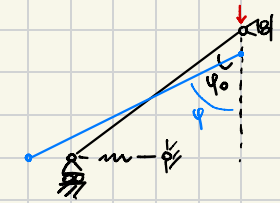
$$\sin \varphi (k l \cos \varphi - P)$$

$$\frac{dV}{d\varphi} = \frac{1}{2} k l^2 \sin 2\varphi - P l \sin \varphi$$

$$= k l^2 \cos 2\varphi - P l \cos \varphi$$

$$\varphi = 0 \quad \forall P$$

$$\varphi \neq 0 \rightarrow P = k l \cos \varphi$$



$$\frac{1}{2} k (l \sin \varphi)^2 + P(l \cos \varphi - l \cos \varphi_0)$$

$$\frac{1}{2} k (l \sin \varphi - l \sin \varphi_0)^2 + P l \cos \varphi - P l \cos \varphi_0$$

$$\frac{dV}{d\varphi} = k l^2 (\sin \varphi - \sin \varphi_0) \cos \varphi - P l \sin \varphi$$

$$P = k l^2 (\sin \varphi - \sin \varphi_0) \cot \varphi$$

$$= k l^2 (\cos \varphi - \sin \varphi_0 \cot \varphi)$$

$$1 + \frac{\alpha^2}{2} - \frac{\sin \varphi_0}{\alpha} \left(\frac{1}{\alpha} \right)$$

$$= \frac{\alpha + \alpha^3 - \sin \varphi_0}{\alpha}$$

$$\frac{d^2V}{d\varphi^2} = k l^2 (\cos^2 \varphi - \sin^2 \varphi + \sin \varphi \sin \varphi_0) - P l \cos \varphi$$

$$= k l^2 (\cos 2\varphi + \sin \varphi \sin \varphi_0) - k l^2 \cos \varphi (\cos \varphi - \sin \varphi_0 \cot \varphi)$$

$$k l^2 \cos 2\varphi - k l^2 \cos^2 \varphi + \sin \varphi \sin \varphi_0 \frac{\cos^2 \varphi}{\sin \varphi}$$

$$k l^2 \left(-\sin^2 \varphi + \sin \varphi_0 \frac{\cos^2 \varphi}{\sin \varphi} + \sin \varphi \sin \varphi_0 \right)$$

$$= \frac{-\sin^3 \varphi + \sin \varphi_0}{\sin \varphi}$$

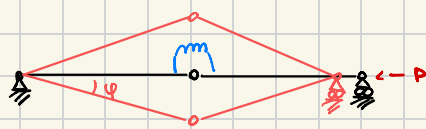
$$= -\sin^2 \varphi + \frac{\sin \varphi_0}{\sin \varphi} > 0$$

$$\boxed{\sin^3 \varphi < \sin \varphi_0}$$

Condizione

$$\varphi^3 < \varphi_0$$

Ex 5:



$$1 - \varphi \cot \varphi$$

$$d < \tan d$$

$$V_{\text{el}} = \frac{1}{2} k (z \varphi)^2$$

$$V(\varphi) = 2k\varphi^2 + 2P\ell(\cos\varphi - 1)$$

$$W_{\text{ext}} = -P\ell\cos\varphi - z\ell$$

$$\frac{dV}{d\varphi} = 4k\varphi - 2P\ell\sin\varphi \sim 0$$

$$P = 2 \frac{k}{\ell} \cdot \frac{\varphi}{\sin\varphi}$$

Stability:

$$\frac{d^2V}{d\varphi^2} = 4k - 2P\ell\cos\varphi$$

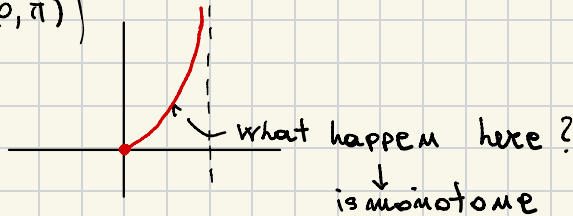
$$4k - 2P\ell\cos\varphi > 0 \text{ (substitute)} \sim 4k - 4k\varphi\cot\varphi > 0$$

Try to plot $(1 - \varphi \cot \varphi)$

1st: symmetric (study only $(0, \pi)$)

$$2^{\text{nd}}: \lim_{\varphi \rightarrow 0} 1 - \frac{\varphi}{\sin\varphi} \cos\varphi = 0$$

$$\lim_{\varphi \rightarrow \pi} 1 - \frac{\varphi}{\sin\varphi} \cos\varphi = +\infty$$



what happens here?
is monotone

$$\frac{d}{d\varphi} = -\cot\varphi - \varphi \cdot \frac{-\sin^2\varphi - \cos^2\varphi}{\sin^2\varphi}$$

So for P_{eq} is always stable

$$-\cot\varphi + \frac{\varphi}{\sin^2\varphi} > 0$$

$$\text{if } \varphi = 0? \quad 4k\ell - 2P\ell > 0 \rightarrow P < 2 \frac{k}{\ell} (= P_{\text{cr}})$$

$$\frac{-\sin\varphi\cos\varphi + \varphi}{\sin^2\varphi} > 0 \sim \varphi > \frac{1}{2}\pi \sin 2\varphi \text{ (always true)}$$

Try to plot P :

$$P = 2 \frac{k}{\ell} \cdot \frac{\varphi}{\sin\varphi}$$

1st: symmetric $(0, \pi)$

$$2^{\text{nd}}: \lim_{\varphi \rightarrow 0} 2 \frac{k}{\ell} \cdot \frac{\varphi}{\sin\varphi} = P_{\text{cr}}$$

$$\lim_{\varphi \rightarrow \pi} 2 \frac{k}{\ell} \cdot \frac{\varphi}{\sin\varphi} = +\infty$$

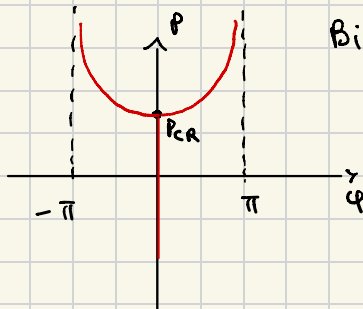
like before

$$\frac{d}{d\varphi} = 2 \frac{k}{\ell} \left(\frac{\sin\varphi - \varphi \cos\varphi}{\sin^2\varphi} \right)$$

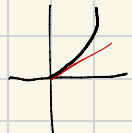
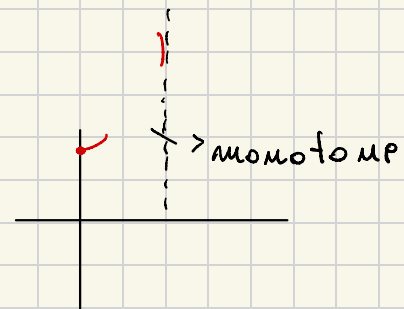
$$\sin\varphi > \varphi \cos\varphi$$

$\varphi < \tan\varphi$ in $(0, \pi)$ is true

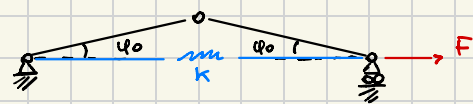
(we know this because $\tan\varphi \approx \varphi$ near 0)



Bifurcation



EX 3:



$$V_{el}(\varphi) = \frac{1}{2} k (z l \cos \varphi - z l \cos \varphi_0)^2$$

$$W_{ext}(\varphi) = P l (z \cos \varphi - z \cos \varphi_0)$$

$$V(\varphi) = z k l^2 (\cos \varphi - \cos \varphi_0)^2 - z P l (\cos \varphi - \cos \varphi_0)$$

$$\frac{dV}{d\varphi} = 4 k l^2 (\cos \varphi - \cos \varphi_0) (-\sin \varphi) - z P l (-\sin \varphi)$$

$$= (-\sin \varphi) (4 k l^2 (\cos \varphi - \cos \varphi_0) - z P l)$$

$$\varphi = 0 \text{ (Stability)} \quad \text{or} \quad P = z k l (\cos \varphi - \cos \varphi_0)$$

$$\frac{d^2V}{d\varphi^2} = (-\cos \varphi) (4 k l^2 (\cos \varphi - \cos \varphi_0) - z P l) + (-\sin \varphi) (-4 k l^2 \sin \varphi)$$

$$= -4 k l^2 \cos^2 \varphi + 4 k l^2 \cos \varphi_0 \cos \varphi + 4 k l^2 \sin^2 \varphi + z P l \cos \varphi$$

$$= -4 k l^2 \cos^2 \varphi + 4 k l^2 \cos \varphi_0 \cos \varphi + 4 k l^2 \sin^2 \varphi + 4 k l^2 \cos^2 \varphi - 4 k l^2 \cos \varphi \cos \varphi_0$$

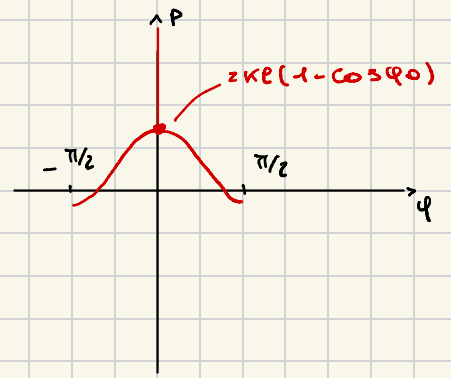
Stability:

$$1 - \cos^2 \varphi > 0 \text{ (Always)}$$

$$\frac{d^2V(0)}{d\varphi} = -4 (4 k l^2 (1 - \cos \varphi_0) - z P l) \text{ (from } \sin \varphi = 0)$$

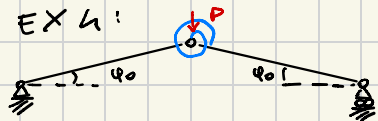
$$4 k l^2 (1 - \cos \varphi_0) - z P l < 0$$

$$P > z k l (1 - \cos \varphi_0)$$



$$\sin^2 \varphi > 0$$

EX 4:



$$V_{el}(\varphi) = \frac{1}{2} k (2\varphi - 2\varphi_0)^2$$

$$W_{ext}(\varphi) = -P l (\sin \varphi - \sin \varphi_0)$$

$$V(\varphi) = 2k(\varphi - \varphi_0)^2 + P l (\sin \varphi - \sin \varphi_0)$$

$$\frac{dV}{d\varphi} = 4k(\varphi - \varphi_0) + P l \cos \varphi \quad \leadsto \quad P = 4 \frac{k}{l} \frac{\varphi_0 - \varphi}{\cos \varphi}$$

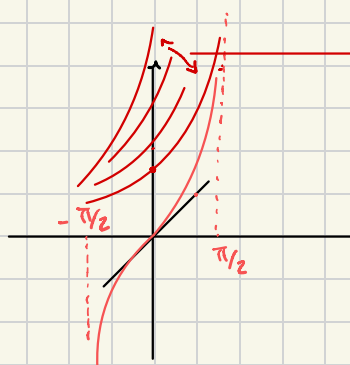
$$\frac{d^2 V}{d\varphi^2} = 4k - P l \sin \varphi$$

$$= 4k - 4k(\varphi_0 - \varphi) + g \varphi$$

$$\lim_{\varphi \rightarrow \pi/2} = -\infty$$

$$\lim_{\varphi \rightarrow -\pi/2} = +\infty$$

We study for $\varphi > 0$ ($\varphi < 0$ equal but you have to change sign) $\leadsto \tan \varphi < \frac{1}{\varphi_0 - \varphi}$

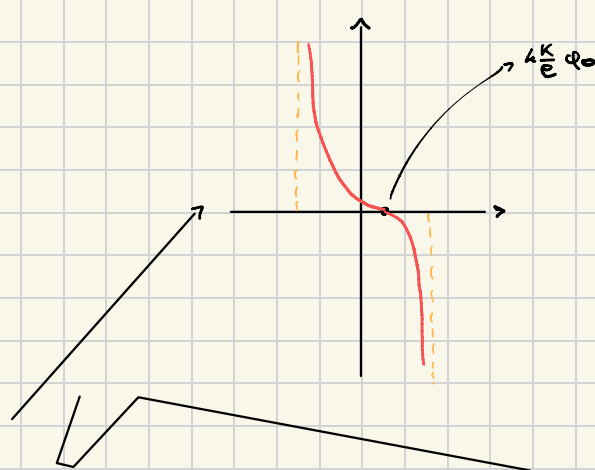


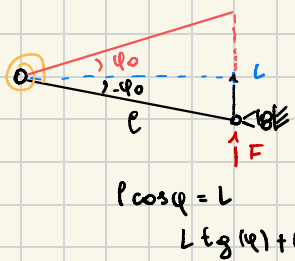
→ This are $\frac{1}{-\varphi}$ and $\frac{1}{\frac{\pi}{2} - \varphi}$ all the other curve for $\varphi_0 \in (0, \pi/2)$

Are between so we conclude that the condition is always true for $\varphi \in (0, \pi/2)$ (also for $\varphi \in (-\pi/2, 0)$ but you would have:

$$\tan \varphi > \frac{1}{\varphi_0 - \varphi})$$

So Always stable





→ 3 map thooog for $\varphi < \frac{\pi}{2}$ (?)

$$V_{el}(\varphi) = \frac{1}{2} k (\varphi + \varphi_0)^2$$

$$W_{ext} = -FL(\tan \varphi + \tan \varphi_0)$$

$$V(\varphi) = \frac{1}{2} k (\varphi + \varphi_0)^2 - FL(\tan \varphi + \tan \varphi_0)$$

$$l \cos \varphi = L$$

$$L \tan(\varphi) + L \tan(\varphi_0)$$

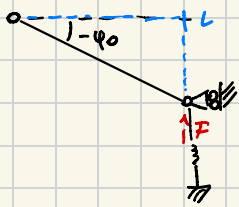
$$F = \frac{k}{e} (\varphi + \varphi_0) \cos^2 \varphi$$

$$\text{For small } \varphi \rightsquigarrow \frac{k}{e} (\varphi + \varphi_0) \left(1 - \frac{\varphi^2}{2}\right) = F$$



$$\varphi + \varphi_0 - \frac{\varphi^3}{2} - \varphi_0 \frac{\varphi^2}{2} = F$$

$$dF = 1 - \frac{3}{2} \varphi^2 - \varphi_0 \varphi < 0$$



$$V(\alpha) = \frac{1}{2} k (L \tan \varphi + L \tan \varphi_0)^2 - FL(\tan \varphi + \tan \varphi_0)$$

$$\frac{dV}{d\alpha} = k L^2 (\tan \varphi + \tan \varphi_0) \cdot \frac{1}{\cos^2 \varphi} - FL \cdot \frac{1}{\cos^2 \varphi}$$

$$\frac{1}{\cos^2 \varphi} (k L^2 (\tan \varphi + \tan \varphi_0) - FL)$$

$$\varphi = \pm \frac{\pi}{2} \text{ not defined}$$

$$F = k L (\tan \varphi + \tan \varphi_0)$$

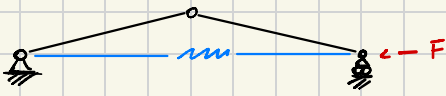
Always stable?

$$-\frac{2 \sin \varphi \cos^3 \varphi}{\cos^4 \varphi} \left(k L^2 \left(\frac{\sin \varphi}{\cos^3 \varphi} + \tan \varphi_0 \right) - FL \right) + \frac{1}{\cos^2 \varphi} \cdot \frac{k L}{\cos^2 \varphi}$$

$\underset{K}{\quad} \quad \quad \underset{0 \text{ (?)}}{\quad} \quad \quad \underset{\frac{k L}{\cos^4 \varphi}}{\quad}$



VAR (come altro ma
grafico inventito)



$$V(\alpha) = 2k\ell^2(\cos\varphi - \cos\varphi_0)^2 + 2F\ell(\cos\varphi - \cos\varphi_0)$$

$$\frac{dV}{d\alpha} = 4k\ell^2(\cos\varphi - \cos\varphi_0)(-\sin\varphi) - 2F\ell\sin\varphi$$

$$(-\sin\varphi)(4k\ell^2(\cos\varphi - \cos\varphi_0) + 2F\ell)$$

$$F = -2k\ell(\cos\varphi - \cos\varphi_0) \quad \forall \varphi = 0 \quad \forall F$$

$$\frac{d^2V}{d\alpha^2} = (-\cos\varphi)(4k\ell^2(\cos\varphi - \cos\varphi_0) + 2F\ell) + (-\sin\varphi)(-4k\ell^2\sin\varphi)$$

$$-4k\ell^2\cos^2\varphi + 4k\ell^2\cancel{\cos\varphi_0\cos\varphi} - 2F\ell\cos\varphi + 4k\ell^2\sin^2\varphi$$

$$+ 4k\ell^2\cos^2\varphi - 4k\ell^2\cancel{\cos\varphi\cos\varphi_0}$$

$$1 - \cos^2\varphi > 0 \quad \text{always}$$

$$-1(4k\ell^2(1 - \cos^2\varphi) + 2F\ell)$$

$$F > 2k\ell(\cos\varphi_0 - 1)$$

$$F < 2k\ell(1 - \cos\varphi_0)$$

